

1D Hydrodynamic Evolution

Using the Finite Volume Method

TASKS

1. Implement a 1D Hydrodynamic Solver

Write a program to solve the 1D Euler equations using the Finite Volume Method (FVM) with the Lax–Friedrichs numerical scheme. The code must evolve the conserved quantities over a 1D spatial grid and accept initial conditions as user input.

2. Sod Shock Tube Test

Validate your solver using the classical Sod Shock Tube problem. Set appropriate left and right initial conditions so that the simulation correctly captures:

- Shock wave
- Contact discontinuity
- Rarefaction wave

REQUIRED OUTPUTS

Generate the following plots as functions of distance along the tube:

1. Density vs Distance
2. Velocity vs Distance
3. Pressure vs Distance

Plots must clearly show the structure of the shock tube solution.

CONCEPTS TO UNDERSTAND

Numerical Methods

- Finite Volume Method (FVM)
- Lax–Friedrichs Method

Stability and Boundary Conditions

- CFL (Courant–Friedrichs–Lewy) Condition
- Boundary conditions in hydrodynamic simulations

RECOMMENDED LANGUAGES

- Python